

MAT 1341A – Test 4

30-November, 2015.

Instructor: Barry Jessup.

Family Name:_____

First Name:_____

Student number:_____

Your multiple choice answers →	1	
	2	
	3	
For the marker's use only →	4	
	5	
	6	
	[Bonus] 7	
	Total	

PLEASE READ THESE INSTRUCTIONS CAREFULLY.

1. You have 80 minutes to complete this exam.
2. This is a closed book exam, and no notes of any kind are permitted. The use of calculators, cell phones, or similar devices is not permitted. All implanted cyber devices not necessary for life-support must be disabled at the beginning of the exam.
3. Read each question carefully, and **answer all questions in the space provided after each question.** For questions 4 to 7, you may use the backs of pages if necessary, but be sure to indicate to the marker that you have done this.
4. Questions 1 to 3 are multiple choice. These questions are worth 1 point each and no part marks will be given. Please record your answers in the space provided above.
5. Questions 4 - 6 are worth 6 points each, and part marks can be earned in each. **The correct answers here require justification written legibly and logically: you must convince the marker that you know why your solution is correct.** Question 7 is a challenging bonus question and is worth 3 points. It is *much* more difficult to obtain marks in the bonus question, so spend your time accordingly. You can earn 100% without attempting Q.7.
6. Where it is possible to check your work, do so.
7. Good luck! Bonne chance!

1. Let A be a 4×5 matrix. Recall that $\ker A = \{x \in \mathbf{R}^5 \mid Ax = 0\}$ and $\operatorname{col} A = \{Ax \mid x \in \mathbf{R}^5\}$. If $\operatorname{rank} A = 4$, then

A. $\operatorname{col} A = \mathbf{R}^4$ and $\dim \ker A = 1$
B. $\operatorname{col} A = \mathbf{R}^4$ and $\dim \ker A = 4$
C. $\dim \operatorname{col} A = 1$ and $\dim \ker A = 1$
D. $\dim \operatorname{col} A = 1$ and $\ker A = \mathbf{R}^5$
E. $\dim \operatorname{col} A = 4$ and $\ker A = \mathbf{R}^5$
F. $\operatorname{col} A = \{0\}$ and $\dim \ker A = 4$

2. Let $B = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & -1 \\ 1 & 1 & -1 \end{bmatrix}$. The third row of B^{-1} is:

A. $[0 \;\; 1 \;\; -1]$
B. $[-1 \;\; 1 \;\; 0]$
C. $[0 \;\; -1 \;\; 1]$
D. $[1 \;\; -1 \;\; 0]$
E. $[1 \;\; 0 \;\; -1]$
F. B is not invertible.

3. Let $S = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ and consider the subset $W = \{A \in \mathbf{M}_{22} \mid SA = AS\}$. Which one of the following statements is true?

A. W is not a subspace of $\mathbf{M}_{22}$
B. W is a subspace of $\mathbf{M}_{22}$, and $\dim W = 4$
C. W is a subspace of $\mathbf{M}_{22}$, and $\dim W = 3$
D. W is a subspace of $\mathbf{M}_{22}$, and $\dim W = 2$
E. W is a subspace of $\mathbf{M}_{22}$, and $\dim W = 1$
F. W is a subspace of $\mathbf{M}_{22}$, and $\dim W = 0$

4. Let

$$A = \begin{bmatrix} 1 & 1 & 0 & -1 \\ 2 & 2 & -1 & 1 \\ 1 & 1 & -1 & 2 \end{bmatrix}.$$

a) Find a basis for the row space of A.

b) Find a basis for the column space of A.

c) Find a basis for $\ker A = \{x \in \mathbf{R}^4 \mid Ax = 0\}$.

d) Find the dot product of each of the vectors in your basis from (a) with each of the vectors in your basis in (c).

5. Let $U = \text{span}\{(1,0,0,1),(1,0,0,0),(0,1,-1,0)\}$.

a) Use the Gram Schmidt algorithm to find an orthogonal basis of U.

b) Find the best approximation to $(1, -1, 2, -1)$ by vectors in U.

c) Extend your basis in (a) to a basis of $\mathbf{R}^4$.

6(a). State whether the following is true or false, and justify your answer.

(i) If A is an invertible matrix and $AB = 0$ then $B = 0$.

ANSWER

(ii) The columns of a 4 by 3 matrix are always linearly dependent.

ANSWER

6(b). Let A be a $n \times n$ matrix with real entries. Give three additional statements equivalent to

"The system $Ax = 0$ has infinitely many solutions"

(i) In terms of the rank of A:

(ii) In terms of the reduced row echelon form of A:

(iii) In terms of the columns of A:

7. [Bonus/Challenge] Let $W = \text{span}\{(1,0,0,1),(1,0,1,0)\}$. Find a 2 by 4 matrix A such that $\ker A = W$.

(You must justify your answer.)

(You may use this page for rough work or solutions that did not fit on previous pages.)